Adversarial attacks

Weekend 4, Saturday morning

Carlos Cotrini

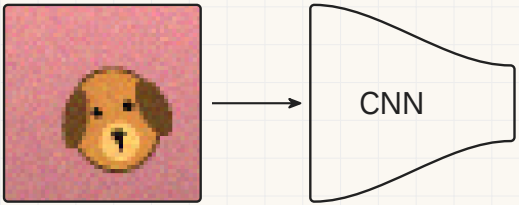
CAS Building ML/AI Applications, ETH Zurich

Saturday 24 October 2026

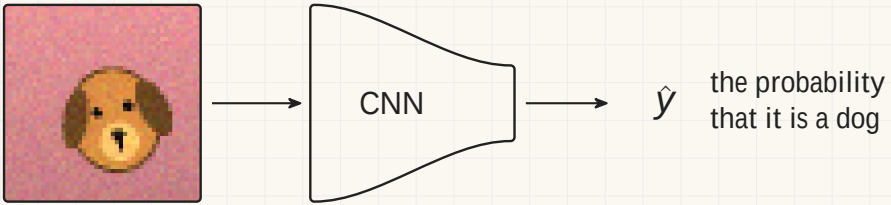
Dog or cat?



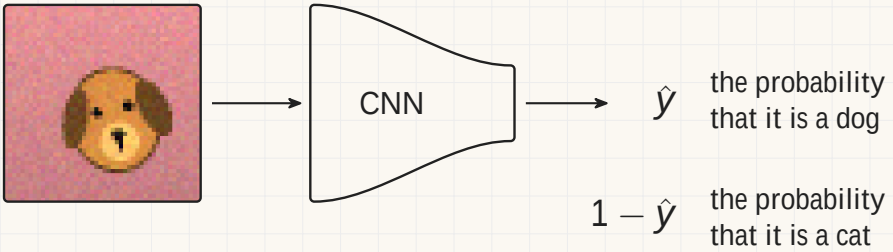
Dog or cat?

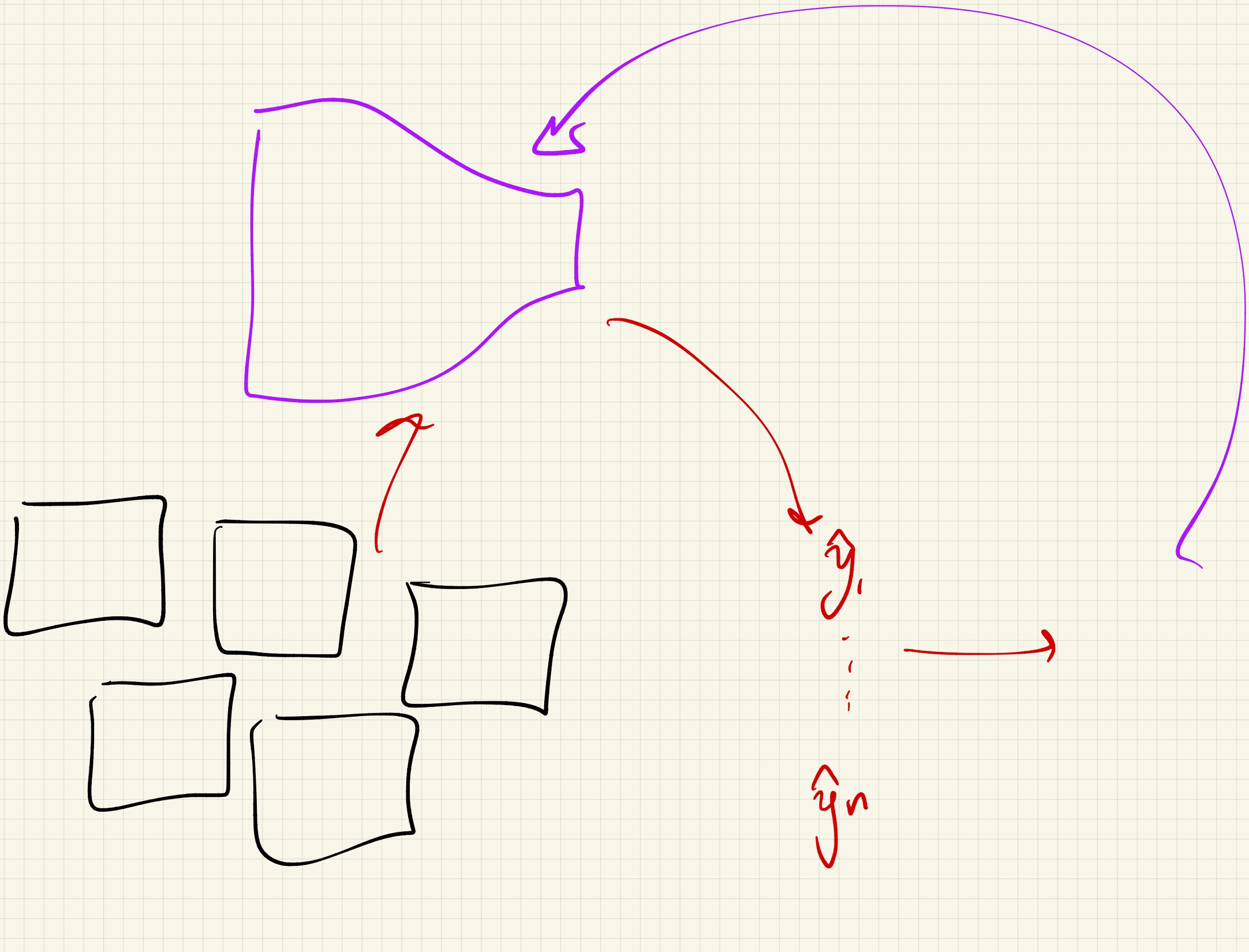


Dog or cat?

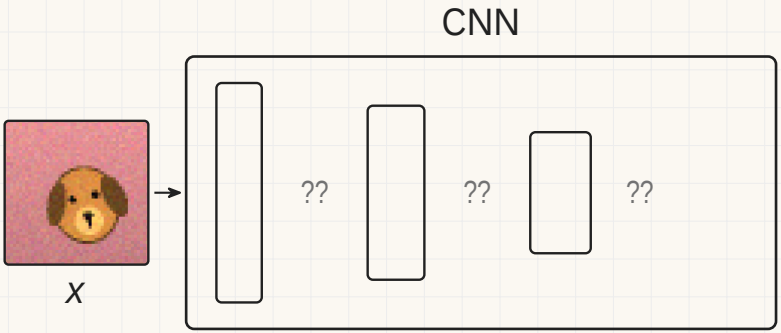


Dog or cat?

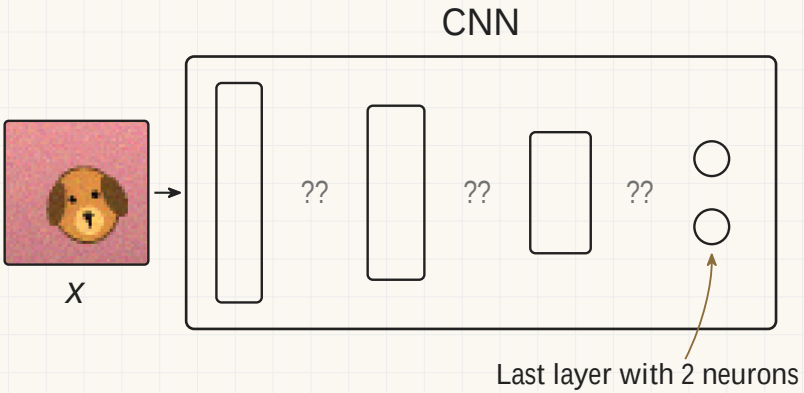

CNN
$\hat{y}$
the probability that it is a dog
$1 - \hat{y}$
the probability that it is a cat


$\hat{y}_1$
$\hat{y}_n$

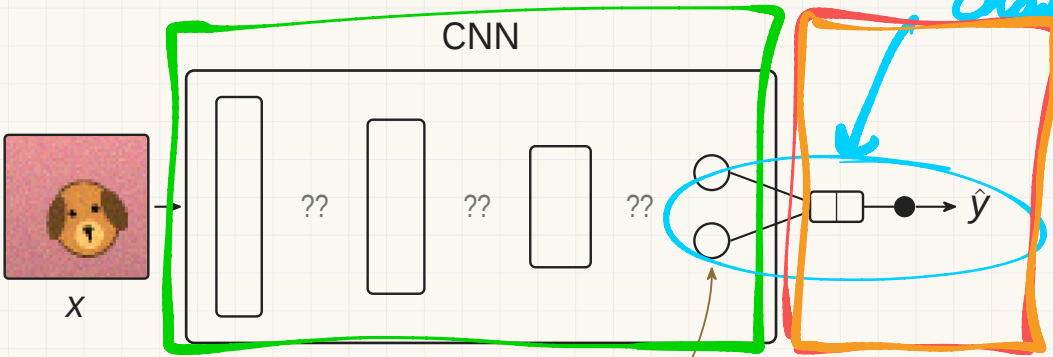
Inside the CNN



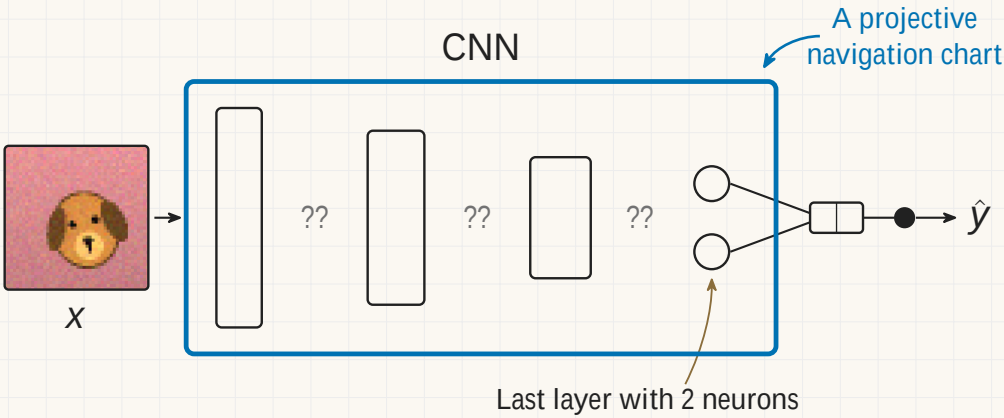
Inside the CNN



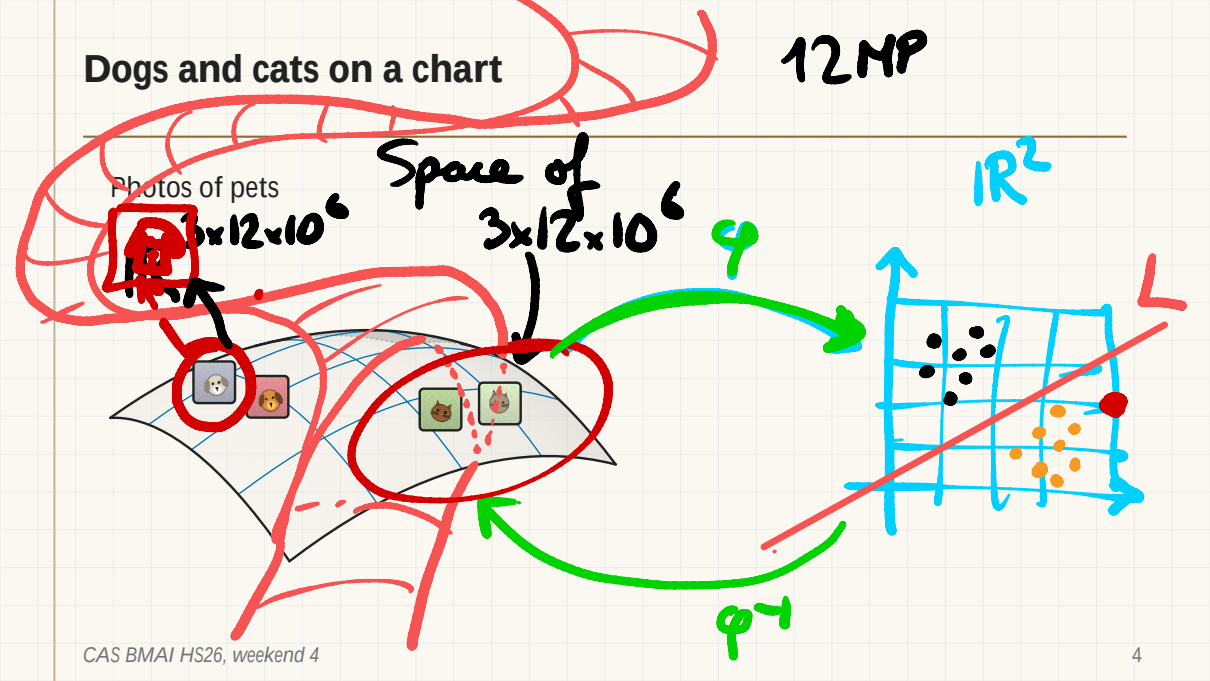
Inside the CNN



Inside the CNN



Dogs and cats on a chart



dog

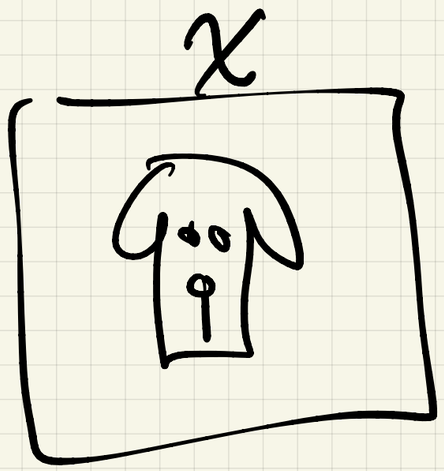
↓

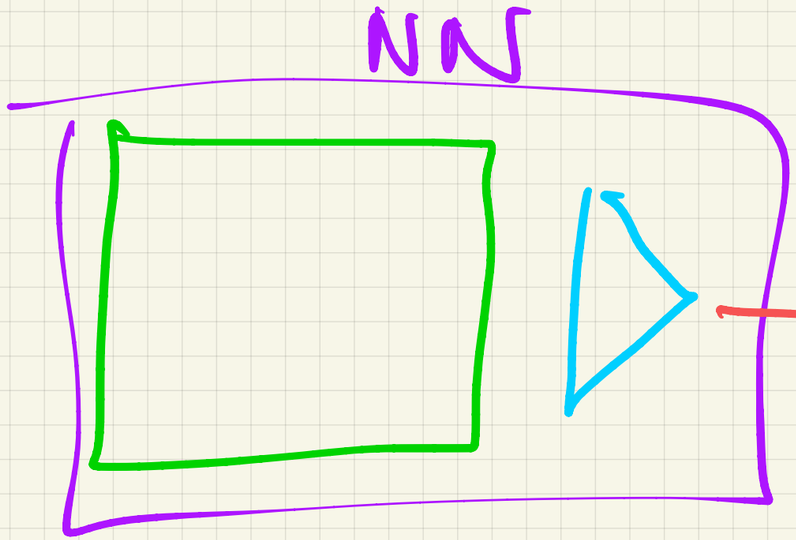
$y = 1$

0

↑

cat





$\hat{y}$

$\mathcal{L}(y, \hat{y})$

↓

$\hat{y} = NN(x)$

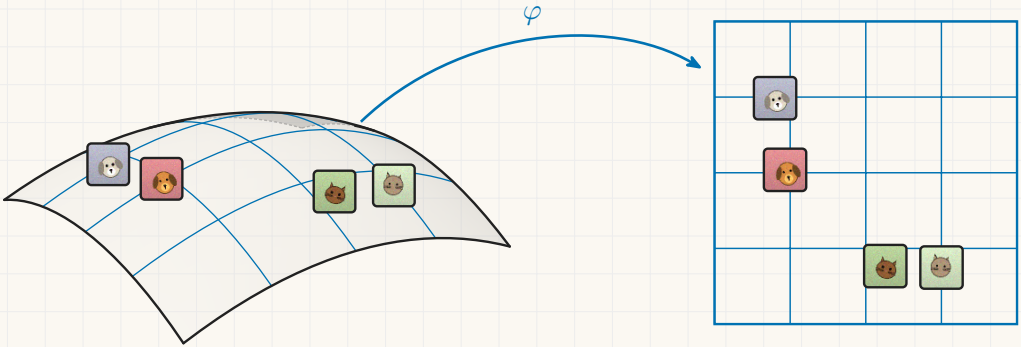
My goal: Change x so that $\hat{y}$ changes

↳ Add ε to x so that it increases $\mathcal{L}(y, NN(x+\varepsilon))$

$$x = x + \varepsilon \nabla_x \mathcal{L}(y, NN(x))$$

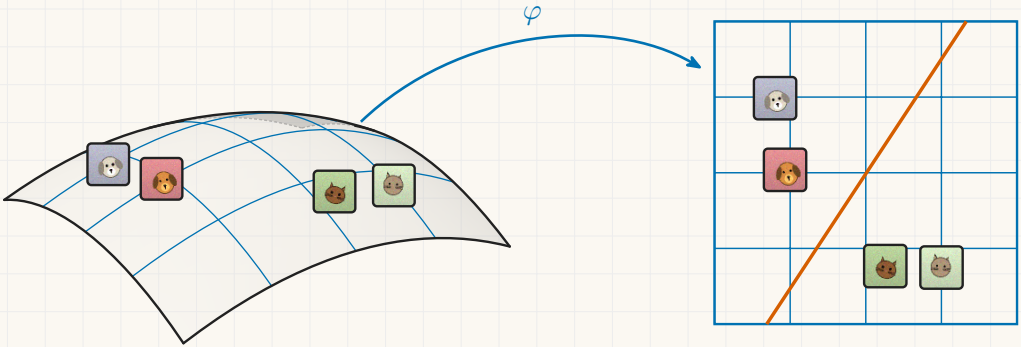
Dogs and cats on a chart

Photos of pets

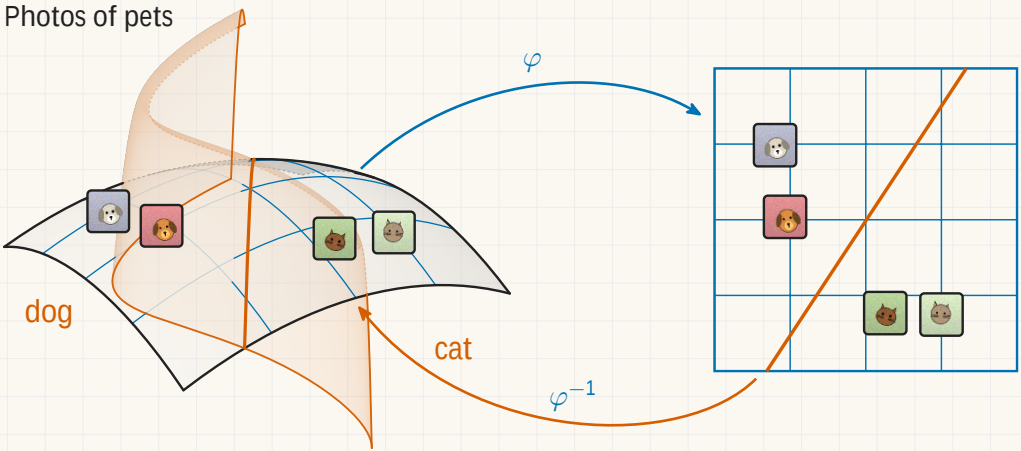


Dogs and cats on a chart

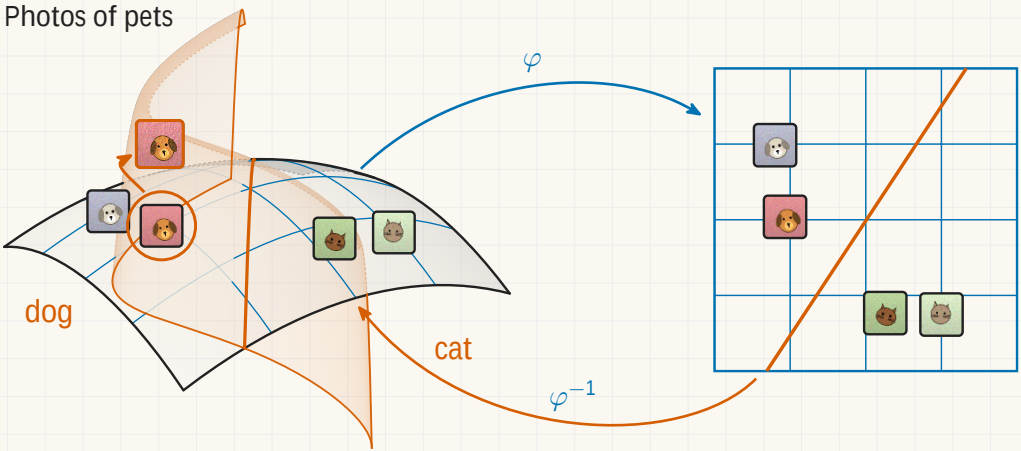
Photos of pets



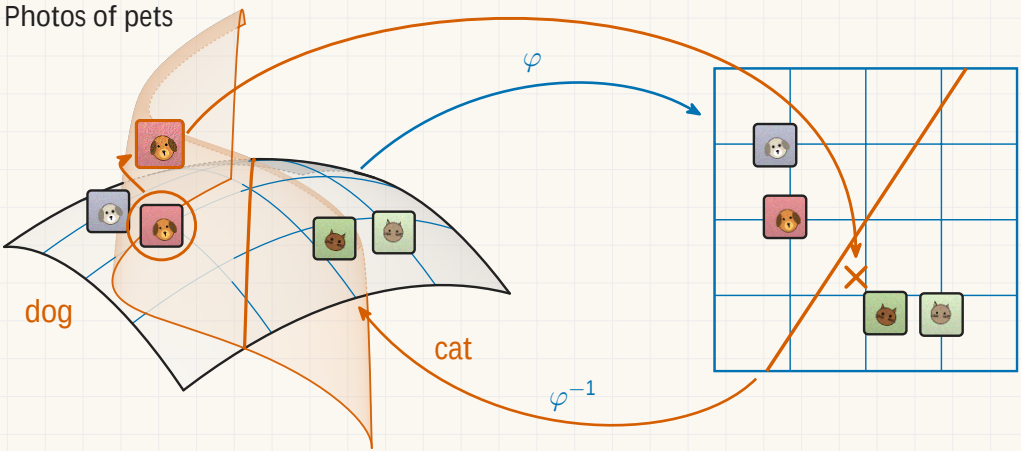
Dogs and cats on a chart



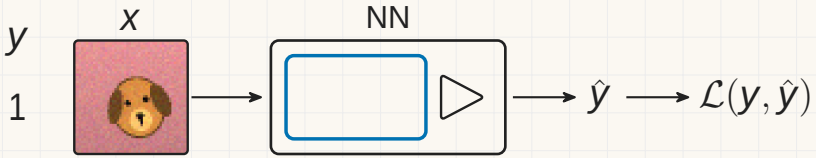
Dogs and cats on a chart



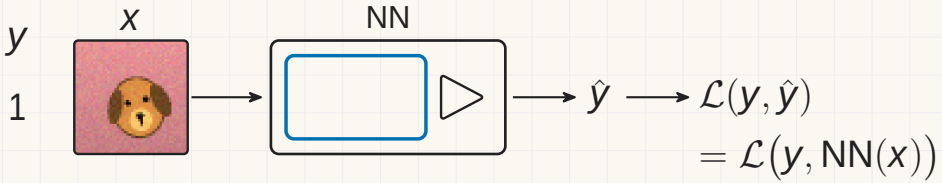
Dogs and cats on a chart



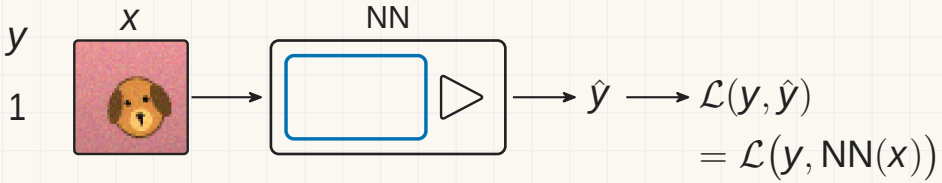
The attack



The attack

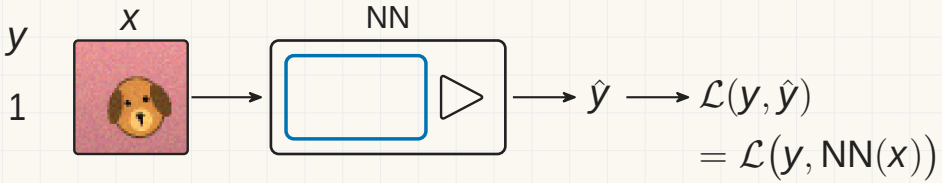


The attack



Goal: change x to change $\hat{y}$.

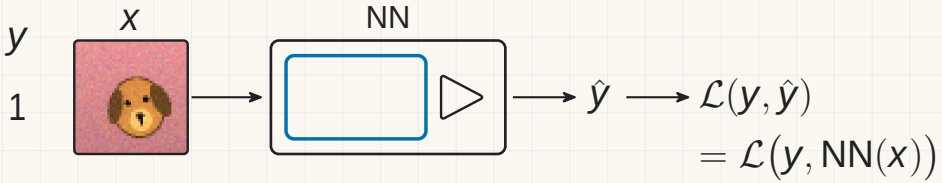
The attack



Goal: change x to change $\hat{y}$.

Add ε to x to increase $\mathcal{L}\big(y, \mathrm{NN}(x)\big)$.

The attack

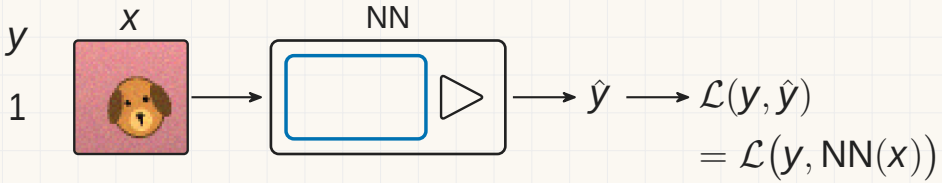


Goal: change x to change $\hat{y}$.

Add ε to x to increase $\mathcal{L}\big(y, \mathrm{NN}(x)\big)$.

$$x = x + \varepsilon \, \nabla_x \, \mathcal{L}\big(y, \mathrm{NN}(x)\big)$$

The attack



Goal: change x to change $\hat{y}$.

Add ε to x to increase $\mathcal{L}\big(y, \mathrm{NN}(x)\big)$.

$$x = x + \varepsilon \, \nabla_x \, \mathcal{L}\big(y, \mathrm{NN}(x)\big)$$

$$\Rightarrow \qquad x = x + \varepsilon \, \mathrm{sgn}\Big(\nabla_x \, \mathcal{L}\big(y, \mathrm{NN}(x)\big)\Big)$$